



COMSATS University Islamabad

Abbottabad Campus

Department of Computer Science

Assignment 03 Spring 2026

Instructor: **Sonia Ghazal**

Subject: **PDC**

Class: **BCS-8B, 8C**

Max Marks: **20**

Instructions:

- Attempt all questions.
- This is a handwritten assignment.
- No printed solutions are allowed.
- Deadline 30/6/2026

CLO-3:

Marks:20

1. Write an OpenMP program to compute the sum of elements of an integer array of size 1000 using the reduction clause. Compare the approach with a sequential implementation.
2. Develop an OpenMP-based program to perform matrix multiplication of two 100×100 matrices using `#pragma omp parallel for`. Explain how loop iterations are distributed among threads.
3. Consider an OpenMP program with 8 iterations executed using 4 threads. Illustrate the execution flow and identify the iterations assigned to each thread under static scheduling.
4. Differentiate between the following OpenMP synchronization mechanisms with suitable examples:
 - Critical
 - Atomic

- Barrier
- Reduction